

Interactive Glassdoor Jobs & Salary Analytics Dashboard

with Machine Learning-Based Salary Prediction

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Problem & Objective

What drives a data job's salary — and can we predict it

The Problem

Job seekers and employers lack clear visibility into what factors actually move salary in data/tech roles. Salary ranges are opaque, inconsistently reported, and rarely broken down by the attributes that matter most — industry, location, required skills, or seniority level.

The Objective

- Analyze real Glassdoor postings to uncover salary drivers across industry, company, location, and required skills
- Build an interactive dashboard so anyone can explore trends visually
- Train an ML model to estimate a fair salary for a given role and company profile

Dataset Overview

GLASSDOOR JOBS DATASET (KAGGLE)

956

Raw Job Postings

Reduced to 467 after removing duplicates and missing salaries

343

Unique Companies

Spanning 60 distinct industries

\$101.5K

Average Salary

Range: \$15K – \$306K across all postings

Fields Used in Analysis

Job Title	Salary Estimate	Rating	Company Size
Revenue	Industry	Location	Job Description

Data Cleaning & Feature Engineering

From raw scrape to model-ready data

Cleaning Steps

→ Salary Parsing

Parsed Salary Estimate text (e.g. "\$53K-\$91K") into numeric `min_salary`, `max_salary`, and `avg_salary` fields

→ Placeholder Removal

Cleaned placeholder values (-1, "Unknown") in Rating, Revenue, Industry, and Size columns

→ Derived Fields

Computed `company_age` from Founded year; extracted `job_state` from Location string

Feature Engineering

7 Skill Flags

Extracted binary flags from job descriptions: **Python, SQL, AWS, Spark, Excel, Tableau, Power BI**

7 Title Categories

Simplified raw job titles into: **Data Scientist, Data Engineer, Analyst, MLE, Manager, Director, Other**

Industry & Location Drive Salary

Highest-paying industries and states

Top-Paying Industries

Computer Hardware & Software

\$116.6K average salary

Biotech & Pharmaceuticals

\$114.0K average salary

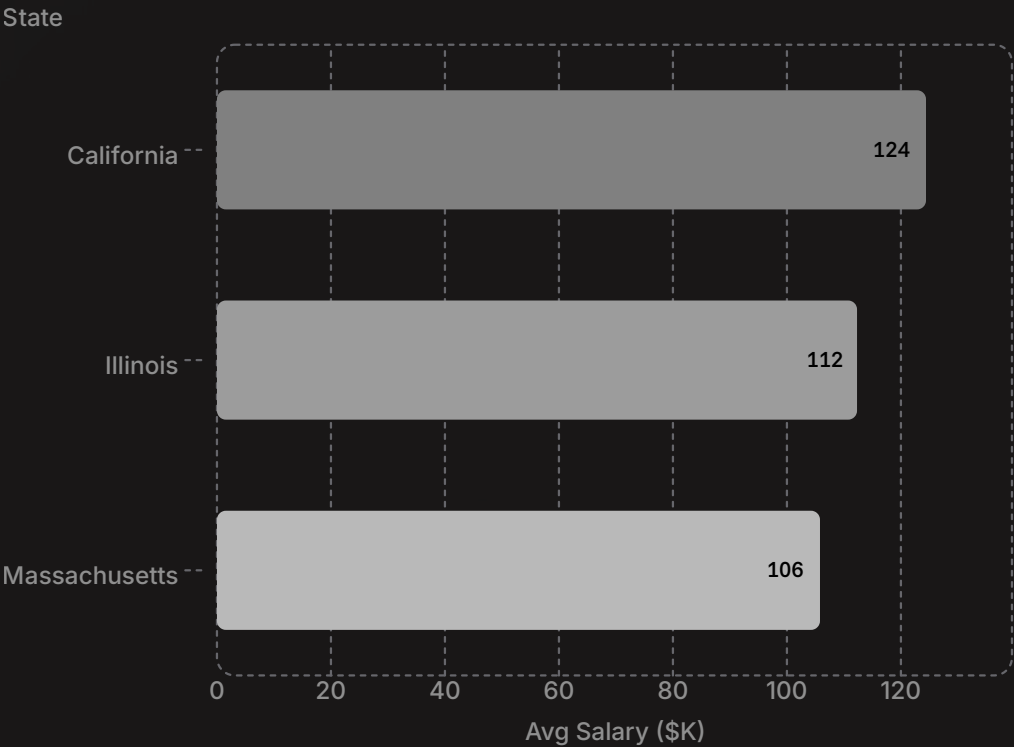
Internet

\$112.8K average salary

Most Job Postings By Industry

Industry	Postings
Biotech & Pharma	62
Insurance Carriers	37
Computer Hardware & Software	37

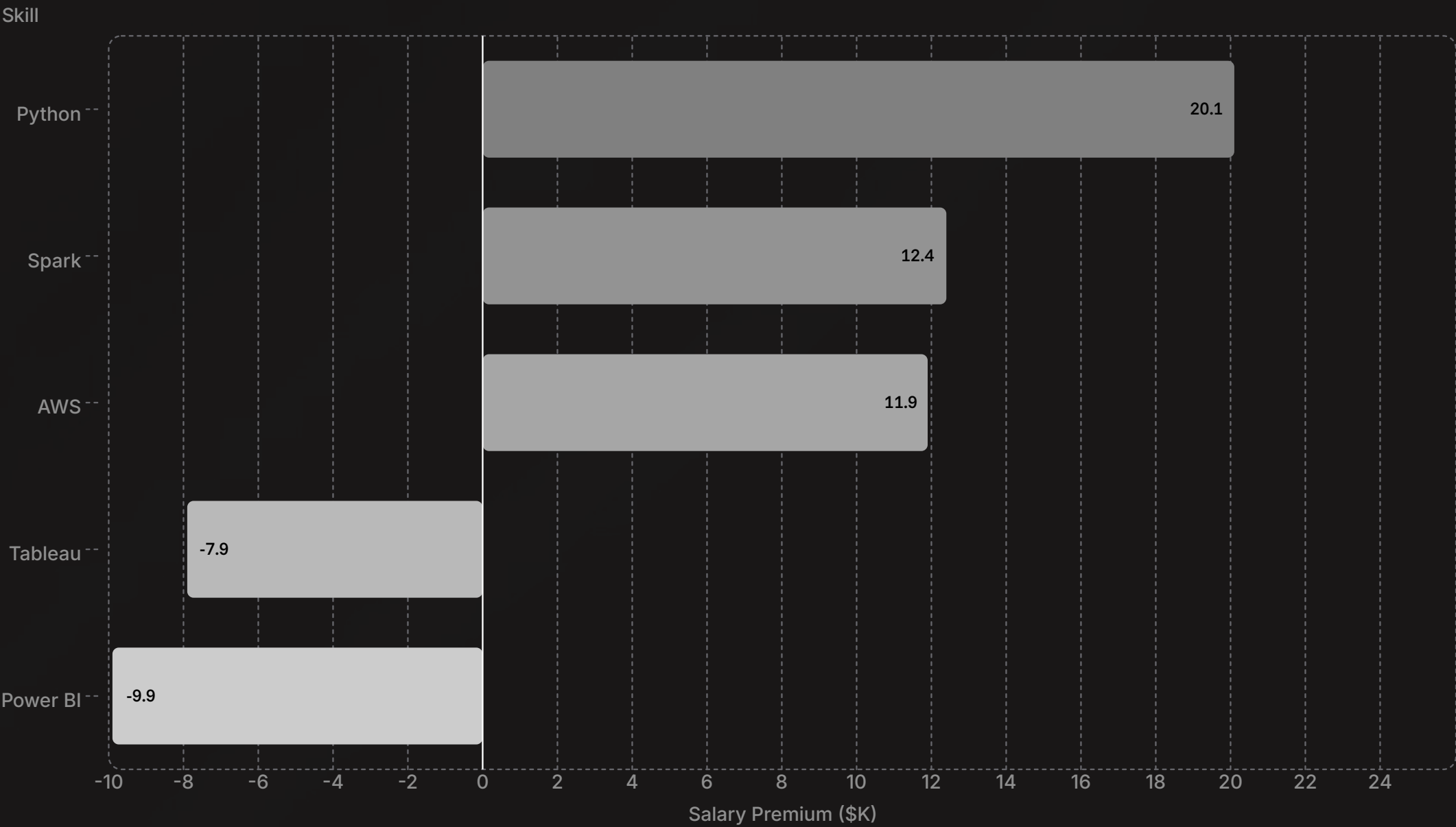
Top-Paying States



California leads significantly at **\$124.4K** average — notably above IL (\$112.3K) and MA (\$105.8K).

Skills vs. Salary

Not all skills pay the same



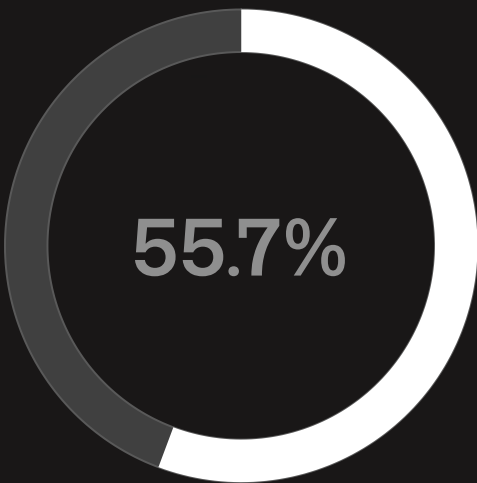
Key Premiums

- **Python** shows the strongest salary premium: +\$20.1K on average (\$110.4K vs \$90.3K without)
- **Spark** (+\$12.4K) and **AWS** (+\$11.9K) also carry clear premiums

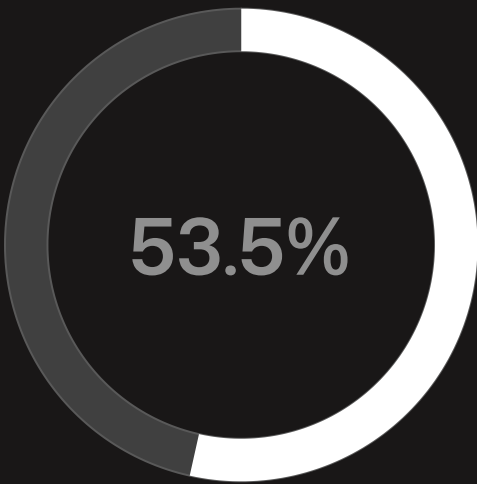
Negative Correlations

- **Tableau** (-\$7.9K) and **Power BI** (-\$9.9K) correlate with lower average salaries — likely because they skew toward analyst-level roles rather than senior data science/engineering roles

Most Requested Skills



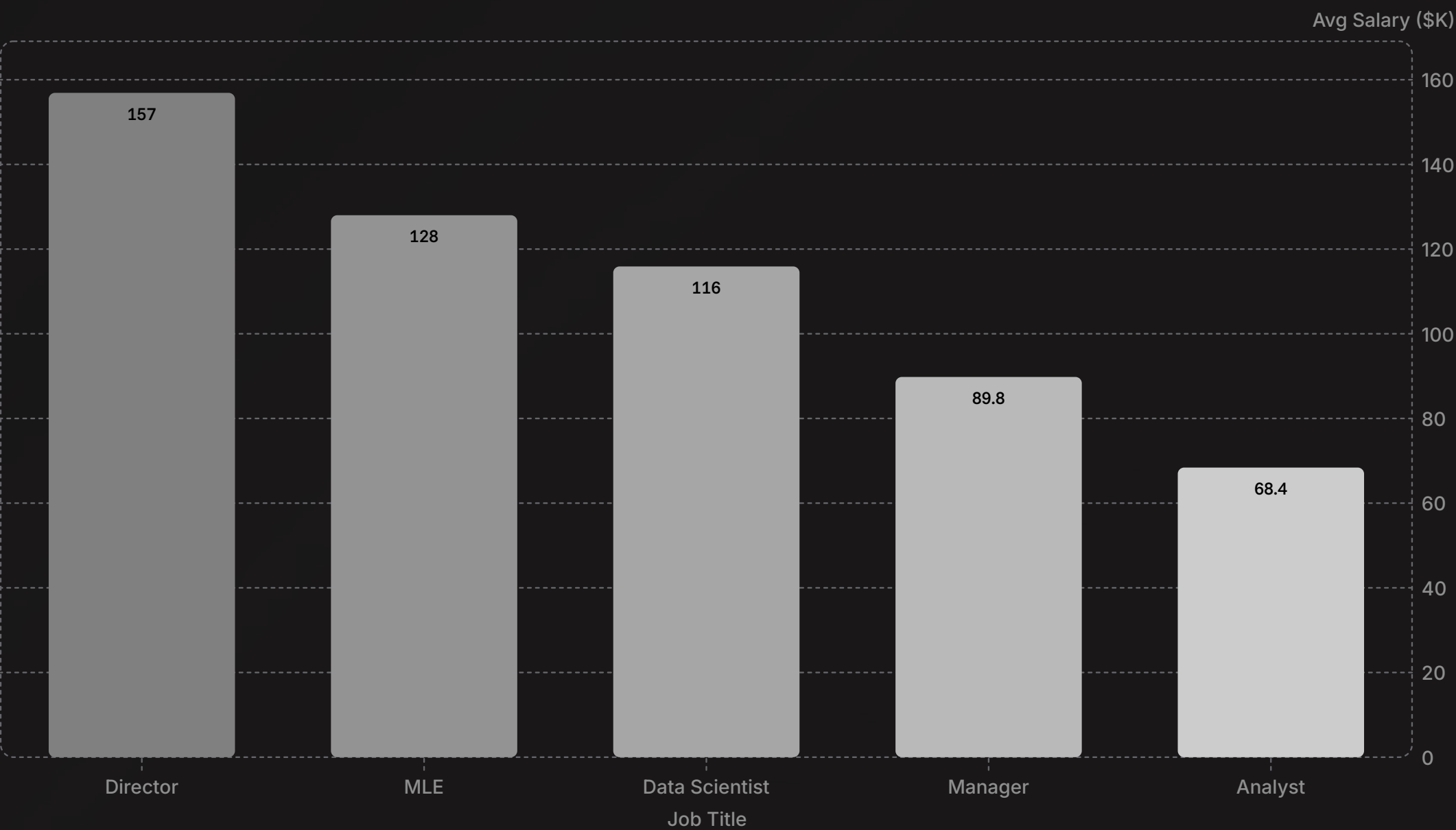
Python



Excel

Job Title & Seniority Matter Most

Role type is the single biggest salary lever



Seniority Gap

Director roles average **\$156.9K** — nearly double an Analyst role at **\$68.4K**. Data Scientist (\$115.9K) and MLE (\$128.0K) roles command strong premiums over Analyst or Manager (\$89.8K) positions.

Company Rating

❑ Company Rating showed only a **weak correlation** with salary ($r = 0.11$) — rating alone doesn't predict pay well.

Machine Learning Model

Predicting salary from job & company attributes

Methodology

01

Model Selection

Compared 5 models: Linear Regression, Decision Tree, Random Forest, Gradient Boosting, XGBoost

02

Evaluation Metrics

Evaluated using MAE, MSE, RMSE, and R^2 on a held-out test set

03

Best Model

Linear Regression selected as best model; saved as `salary_model.pkl`

📌 **Note on modest R^2 :** Reflects the dataset's limited size (467 rows) and high salary variance even within identical job/company profiles — a realistic constraint of small real-world datasets.

Model Comparison: R^2 and RMSE

Model	R^2	RMSE (\$K)
Linear Regression (best)	0.139	37.36
Random Forest	0.131	37.53
Gradient Boosting	0.115	37.86
XGBoost	0.054	39.15
Decision Tree	-0.316	46.18

Live Dashboard Demo

Explore it yourself

 Live Dashboard

<https://data-visualization-major-project-with-app-dashboard-pwwwlvrqkq.streamlit.app/>

 GitHub Repository

<https://github.com/Pradeep1208/Data-Visualization-Major-Project-with-Streamlit-Dashboard>

5 Interactive Pages

Dashboard Overview

Company Analysis

Salary Explorer

Skills Analysis

Salary Predictor



Example Prediction: A senior Data Scientist role at a 4.2-rated private Internet company in CA → predicted **\$142.7K**

Conclusion

Key takeaways



Seniority & Industry Are the Strongest Predictors

Job title/seniority and industry are stronger salary predictors than company rating. Director roles earn nearly double Analyst roles; Computer Hardware & Software, Biotech, and Internet lead on pay.



Python, AWS & Spark Carry the Clearest Skill Premiums

Python commands a +\$20.1K premium on average. Visualization tools (Tableau/Power BI) correlate with lower salaries in this dataset — reflecting their concentration in analyst-level roles.



California Pays Significantly Above Other States

At \$124.4K average, California outpaces Illinois (\$112.3K) and Massachusetts (\$105.8K) for data roles — geography remains a meaningful salary lever.